

Zichun Yu

6413 Gates Hillman Complex, 4902 Forbes Ave, Pittsburgh, PA 15213

✉ zichunyu@andrew.cmu.edu | 🏠 yuzc19.github.io | 🎓 Google Scholar

Education

Carnegie Mellon University (CMU)

PHD STUDENT AT LANGUAGE TECHNOLOGIES INSTITUTE

Aug. 2023 - Present

- GPA: 4.0/4.0, Amazon AI Ph.D. Fellowship Program

Tsinghua University (THU)

B.ENG. IN COMPUTER SCIENCE AND TECHNOLOGY

Sep. 2019 - Jul. 2023

- GPA: 3.94/4.0, Ranking: 7/202

Research Interests

- *Intelligent and efficient LLM scaling with novel pretraining data crawling, refining, selection, and synthesis methods.*
- *Training better AI scientists via adaptive environment generation, synthetic data curation, and self-directed feedback.*

Professional Experience

Meta Platforms

RESEARCH INTERN (WORKING WITH JASON WESTON)

May. 2026 - Present

- Research on thinking-augmented mid-training and agent-environment co-evolution for recursive self-improvement.

RESEARCH INTERN (WORKING WITH SCOTT YIH)

May. - Oct. 2024

- Research on group-level data selection methods for efficient pretraining; first-author paper published at NeurIPS 2025.

Selected Works

- **Targeted Web Crawling for Efficient Pretraining Data Acquisition** (Ongoing)

Zichun Yu, Zhihan Zhang, Tony Li, Sai Gopal, Chenyan Xiong

TL;DR: Predicting outlink priority via inlink document features, improving web crawling efficiency and pretraining data quality.

- **ReScraper: Unified Scraping and Cleaning of Web Data for Effective LLM Pretraining** (Under Review) [PDF]

Zichun Yu*, Jiarui Yan*, Shlok Sanghvi, Nihar Atri, Chenyan Xiong

TL;DR: A 0.6B model that replaces heuristic scraping and cleaning, turning web data into pretraining-ready text in one pass.

- **Generating Pretraining Tokens from Organic Data for Data-Bound Scaling** (COLM'26) [PDF]

Zichun Yu, Chenyan Xiong

TL;DR: A model-aware data synthesis framework that helps LLMs learn more from organic data in the data-bound regime.

- **RePro: Training Language Models to Faithfully Recycle the Web for Pretraining** (ICML'26) [PDF]

Zichun Yu, Chenyan Xiong

TL;DR: Training a small LM with carefully designed RL to faithfully rephrase web text into higher-quality pretraining data.

- **Group-Level Data Selection for Efficient Pretraining** (NeurIPS'25) [PDF]

Zichun Yu, Fei Peng, Jie Lei, Arnold Overwijk, Wen-tau Yih, Chenyan Xiong

TL;DR: Group-level data selection method that improves pretraining efficiency by modeling interactions among training data.

- **Montessori-Instruct: Generate Influential Training Data Tailored for Student Learning** (ICLR'25) [PDF]

Xiaochuan Li, Zichun Yu, Chenyan Xiong

TL;DR: Training a teacher model to generate synthetic data tailored to a student model's learning through data influence.

- **MATES: Model-Aware Data Selection for Efficient Pretraining with Data Influence Models** (NeurIPS'24) [PDF]

Zichun Yu, Spandan Das, Chenyan Xiong

TL;DR: Model-aware data selection that trains a small data influence model to pick influential pretraining examples on the fly.

- **Augmentation-Adapted Retriever Improves Generalization of Language Models as Generic Plug-In** (ACL'23) [PDF]

Zichun Yu, Chenyan Xiong, Shi Yu, Zhiyuan Liu

TL;DR: Adapting a retriever based on LM preferences on documents to improve LM zero-shot generalization on knowledge tasks.